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PRESTON (PO-YEN) TSENG

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Professional Summary

Hands-on Tech Lead with 7+ years of experience building cloud and enterprise systems. Focused on cloud platforms for safety-critical autonomous rail operations since 2024. Lead a 3-5 engineer team, own system architecture, and manage roadmap delivery. Build systems that are easier to monitor, test, and automate.

Technical Skills

Languages: Python, TypeScript, Go, C#

Backend, Data & Messaging: FastAPI, GraphQL, SQLAlchemy, asyncio, ASP.NET Core, PostgreSQL, TimescaleDB, MySQL, MongoDB, MQTT

Frontend: Angular, Vue, Nuxt.js, D3.js, PrimeNG, Vuetify

Platform & Quality: Distributed systems, Docker, Jenkins, Azure DevOps, Playwright, Grafana, CI/CD, E2E automation, event recording and replay

AI & Domains: Hermes Agent, Azure AI Service, OpenAI GPT, Semantic Kernel, agent-assisted workflows, safety-critical autonomous rail, timetabling, mission execution, FSM design, BIM automation

Experience

Tech Lead, LILEE Systems – New Taipei City, Taiwan — Apr 2026 – present

- Lead a 3-5 engineer cloud team responsible for timetabling, mission management, safety-control workflows, and observability. Own architecture across services and work with the CEO on roadmaps and KPIs.
- Built Hermes Agent workflows that turn engineering specifications into delivery plans and verified Confluence and Jira updates. Created and shared reusable LILEE Systems and railway-rule skills for the team.
- Built a human-reviewed, agent-assisted hiring workflow used for 13 candidates. It replaced repeated PowerPoint and Excel handoffs with traceable Confluence records and reduced HR coordination work.

Full Stack Engineer, LILEE Systems – New Taipei City, Taiwan — May 2024 – Mar 2026

- Designed and built the timetable data model, APIs, administration, and live status views. The reviewed API stayed stable for later station- and vehicle-centric views, supporting a recurring 1,351-service weekend testbed schedule with 1–3-second full-day conflict detection.
- Redesigned the cloud-side Safety Server into stateful Core Services and Mission Execution layers with explicit mission and Movement Authority lifecycles. Grafana made hidden testbed failures traceable within seconds.
- Built Faramund, a testbed dispatcher agent that opens wayside signals, guides a driver agent with a failed vehicle back to the yard, resolves incidents, and alerts the team without continuous monitoring.
- Built a Go and TimescaleDB MQTT recorder and replay tool that helps engineers reconstruct cross-service incidents without reproducing them live.
- Built the Jenkins release gate for teammate-implemented Playwright tests and managed the team-authored regression plan. It first caught 1-2 bugs per build and helped establish pre-push testing; release findings became rare within a month.

AI Technical Lead, WeBIM Services – Taipei, Taiwan — Jan 2024 – May 2024

- Led enterprise AI and frontend migration initiatives using Azure AI, OpenAI GPT, Semantic Kernel, and micro-frontend architecture. Covered document search, customer service automation, and Vue2/Vue3 web platform modernization.
- Developed an enterprise knowledge-management system using Azure AI Service to digitize paper documents and improve internal search and retrieval workflows.
- Built an OpenAI GPT-powered customer-service automation system that reduced manual customer-service workload by 86%.

- Introduced a micro-frontend migration architecture that enabled Vue2 and Vue3 services to coexist during frontend modernization.

Full Stack Engineer, WeBIM Services – Taipei, Taiwan July 2019 – Jan 2024

- Built SyncoBox Automation to automate the end-to-end BIM model inspection workflow, reducing manual review effort for architectural and engineering teams.
- Designed SyncoBox Markups to move paper-based architectural design discussions into online markup and collaboration workflows.
- Developed facility-management modules that streamlined factory maintenance workflows.
- Established Azure-based CI/CD and developed shared backend packages, permission logic, and microservice interfaces, improving maintainability and cross-team collaboration.
- Built computer-vision-based CCTV electronic-fence and alarm features integrated with BIM models to improve construction-site worker safety.

Certifications & Languages

Certifications: Microsoft Certified: Azure AI Engineer Associate

Languages: Chinese (native), English (professional working proficiency, TOEFL: 93/120), Japanese (JLPT N2)

Education

National Taiwan University, Master's degree in Transportation Engineering 2020 – 2022

- GPA: 3.93/4.3
- Emergency Indoor Positioning System with Visual SLAM to Improve the Efficiency of EMS, SCEM International Conference, 2022

National Taiwan University, Bachelor's degree in Civil Engineering 2016 – 2020

- GPA: 3.92/4.3

Publications

Real-time indoor localization with visual SLAM for in-building emergency response May 2022

Po-Yen Tseng, Jacob J. Lin, Ying-Chieh Chan, Albert Y. Chen

10.1016/j.autcon.2022.104319 (Automation in Construction)